Reflection and Traceability Report on Software Engineering

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1 Changes in Response to Feedback

1.1 Problem Statement, Goals, and Development Plan

Table 1 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 1: Feedback Responses and Traceability for Problem Statement, Goals, Development Plan

Feedback Source	**Summary**	**Changes Made**	**Related Issues**
Peer	Learning outcome extraction lacks a measurable success criteria	Added measurable success criteria to the learning outcome extraction	IS#32, Commit#2a966de
Peer	Word document support is not explicitly addressed in the schedule	Clarified that initially only PDF support will be implemented and Word document support will be documented as future work.	IS#33, Commit#f07d482
TA	Need more details in the team charter: How is "repeatedly misses a commitment" defined? What's the exact amount? What would work balancing look like?	Updated team charter to include examples and more details.	IS#94, PR#117
Peer	The PoC fallback could leave the CP-SAT constraint risk unresolved	Prioritized a reduced CP-SAT model and clarified that any scoring fallback must document the unresolved constraints and evaluate them before full implementation.	IS#35
Peer	CEAB validation is not clearly demonstrated in the PoC	Added one representative CEAB graduate-attribute check to the PoC and clarified that broader CEAB coverage will be evaluated afterward.	IS#34
TA	The PoC plan is too ambitious and does not clearly prioritize its main risks	Made reliable data extraction and alternative scheduling dates the primary PoC focus, while limiting policy, CEAB, explanation, and other features to small demonstrations or follow-up work.	IS#93

Table 1: Feedback Responses and Traceability for Problem Statement, Goals, Development Plan

Feedback Source	Summary	Changes Made	Related Issues
Peer	The PoC integrates too many components (storage, calendar, conflict detection, alternative dates, LLM explanations, compliance checks) to evaluate the main risks before the demonstration	Narrowed the PoC in the Development Plan [1] to one outline going through the extraction pipeline: PDF upload, PyMuPDF text extraction, a local LLM returning schema-checked JSON, and a frontend table showing each value with its source passage. Listed the calendar, conflict detection, CP-SAT suggestions, policy and CEAB checks, LLM explanations, login and export as not part of the demonstration, and gave a fallback for each risk. This replaces the PoC scope described for IS#34, IS#35 and IS#93, since date suggestions became an extra after the meeting with the department.	IS#36, PR#140
TA	Expected Technology needs clearer justification beyond "X tool/technology does Y" (Eval Tools, LO13)	Rewrote the Expected Technology section of the Development Plan [1] so that each tool states how it is used, why it was chosen over the alternatives considered (for example, PyMuPDF over pdfplumber based on the model-selection spike results, Ollama because the model must run on CAS servers at no cost), and its limitations. Added the tools that were missing (PyMuPDF, Ollama, the candidate open-weight models, OR-Tools CP-SAT), and marked vLLM, CUDA, Transformers and PyTorch as conditional.	TA rubric feedback, PR#140
TA	Keep the quantified success criteria consistent with later documents such as the SRS	Raised the targets in Goals 2 and 5 of the Problem Statement [2] from 90% to 95% to match NFR1 and Appendix C of the SRS [3], added a 95% target for learning outcomes (with their graduate attributes) to both documents, and changed "Software Engineering" to all CAS undergraduate programs in Goals 1 and 3 to match the SRS cohorts.	IS#91, PR#140

Table 1: Feedback Responses and Traceability for Problem Statement, Goals, Development Plan

Feedback Source	Summary	Changes Made	Related Issues
TA	The predefined outcomes and datasets lack validation; define the ground truth	Added a "Test dataset and ground truth" paragraph to the Goals: real and mock outlines, and an answer key built by the team that lists every expected value, missing field, conflict and compliance issue, double-checked against the outlines before it is used for scoring.	IS#91, PR#140
TA	Has thought been given to courses outside the department (e.g. math, statistics, engineering) that still affect the schedule?	Added a "Courses covered" paragraph to the Goals: required courses from other departments and first-year engineering courses are not modelled because CAS does not receive or control their outlines, so the workload shown is a lower bound. Added the matching assumption A8 to the SRS [3].	IS#91, PR#140

1.2 SRS and Hazard Analysis

The SRS and Hazard Analysis deliverables are closely related, so we will summarize the feedback received for these two deliverables together in the sections that follow.

1.2.1 SRS

Table 2 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 2: Feedback Responses and Traceability for SRS

Feedback Source	Summary	Changes Made	Related Issues
Peer	Session timeout protection is not formally specified in the SRS and Hazard Analysis	Added a requirement specifying that authenticated sessions should expire after a defined period of inactivity in the SRS and updated the Hazard Analysis accordingly	IS#86, PR#118
Peer	Model changes could reduce extraction accuracy without a formal regression requirement	Added a requirement to rerun the representative accuracy dataset and confirm the thresholds before deploying a new or updated model.	IS#84
Peer	CSV export did not specify protection against spreadsheet formula interpretation	Updated the CSV export requirement to escape values beginning with formula characters.	IS#85

Table 2: Feedback Responses and Traceability for SRS

Feedback Source	Summary	Changes Made	Related Issues
Peer	The SRS does not say how model outputs that fail validation or contain unexpected instructions are rejected (prompt injection)	Added AIR7 (output validation) to the SRS [3]: outline text is given to the model only as data and output is never acted on as an instruction; malformed output is rejected and the outline flagged for manual entry; values without a matching source passage or out of range are treated as missing; policy violations are not rejected; a model is only used if its validated output is unchanged on the prompt-injection test set. Also added a prompt-injection row to Known Limitations, a definition of prompt injection, a rationale item, and AIR7 to the traceability tables.	IS#82, PR#140
TA	Keep the success criteria consistent between the Problem Statement and the SRS; consider courses outside the department	Added the 95% learning outcome accuracy target to the SRS [3] (NFR1, Evaluation Metrics, *LO_ACCURACY* in Appendix C), changed the source of the accuracy constants to the goals in the Problem Statement [2], updated the accuracy rationale, stated that the answer key is double-checked, and added assumption A8 (only CAS courses are modelled).	IS#91, PR#140

1.2.2 Hazard Analysis

Table 3 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 3: Feedback Responses and Traceability for Hazard Analysis

Feedback Source	Summary	Changes Made	Related Issues
Peer	Backup and recovery safeguards are not clearly defined	Clarified H2-1 by assigning backups and recovery to CAS IT, requiring the project team to document application data and verify representative restores, and adding a backup and restore requirement to the roadmap.	IS#83, PR#123, ...

Table 3: Feedback Responses and Traceability for Hazard Analysis

Feedback Source	Summary	Changes Made	Related Issues
Peer	CSV formula injection prevention was not included in the safety requirements	Added SSR_CSVExport to formalize escaping dangerous spreadsheet formula prefixes and testing the resulting export.	IS#85
Peer	Model changes lacked a defined regression testing requirement	Added SSR_ModelChangeTest to require accuracy testing and recording the model version before deployment.	IS#84
Peer	The output validation actions for prompt injection (H4-5) are marked "New" but are not in the Safety and Security Requirements	Added SSR13 (SSR_OutputValidation) to Table 5 of the Hazard Analysis [4] and linked it to AIR7 in the SRS [3]; replaced "New: output validation" in row H4-5 with SSR13 and AIR7; added AIR7 to row H4-1; and listed AIR7 among the SRS safeguards in Section 6. This also responds, for H4-5, to the TA comment that "New" requirements should be added to Table 5.	IS#82, PR#140

1.3 Design and Design Documentation

The design and design documentation deliverables include the software architecture and detailed design. We will summarize the feedback received for each of these deliverables separately in the sections that follow.

1.3.1 Software Architecture

Table 4 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 4: Feedback Responses and Traceability for Software Architecture

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.3.2 Detailed Design

Table 5 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 5: Feedback Responses and Traceability for Detailed Design

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.4 VnV Plan and Report

The VnV Plan and Report deliverables are closely related, so we will summarize the feedback received for these two deliverables together in the sections that follow.

1.4.1 VnV Plan

Table 6 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 6: Feedback Responses and Traceability for VnV Plan

Feedback Source	Summary	Changes Made	Related Issues
Peer	A new output validation requirement (prompt injection) needs to be verified	Added system test ST_OutputValidation to the VnV Plan [5] (injected instructions, malformed output, values without a source passage or out of range) and traced AIR7 and SSR13 to it in the traceability table.	IS#82, PR#140
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

1.4.2 VnV Report

Table 7 summarizes the feedback received from our peers, the TA, the instructor and other sources. The table includes links to the related issues and pull requests, along with a summary of our response/changes.

Table 7: Feedback Responses and Traceability for VnV Report

Feedback Source	Summary	Changes Made	Related Issues
Peer	summary of feedback	summary of changes made	IS#, PR#, ...
TA	summary of feedback	summary of changes made	IS#, PR#, ...
Instructor	summary of feedback	summary of changes made	IS#, PR#, ...
Other	summary of feedback	summary of changes made	IS#, PR#, ...

2 Custom Documents (CDs)

[Summarize the CDs that were tackled by this project. CDs can include usability testing, code walkthroughs, user documentation, formal proof, GenderMag personas, Design Thinking, etc. CDs should have already been

approved by the course instructor as included in your problem statement. The CDs should match what is documented in TeamComposition.csv. —TPLT]

[Provide links to the documentation of the CDs —TPLT]

3 Design Iteration (LO10 (Prototypes))

[Explain how you arrived at your final design and implementation. How did the design evolve from the first version to the final version? —TPLT]

[Don't just say what you changed, say why you changed it. The needs of the client should be part of the explanation. For example, if you made changes in response to usability testing, explain what the testing found and what changes it led to. —TPLT]

4 Design Decisions (LO11) (Justifies)

[Reflect and justify your design decisions. How did limitations, assumptions, and constraints influence your decisions? Discuss each of these separately. —TPLT]

5 Economic Considerations (LO19) (Economic)

[Is there a market for your product? What would be involved in marketing your product? What is your estimate of the cost to produce a version that you could sell? What would you charge for your product? How many units would you have to sell to make money? If your product isn't something that would be sold, like an open source project, how would you go about attracting users? How many potential users currently exist? —TPLT]

6 Reflection on Project Management (LO20) (Manages)

[This question focuses on processes and tools used for project management. —TPLT]

6.1 How Does Your Project Management Compare to Your Development Plan

[Did you follow your Development plan, with respect to the team meeting plan, team communication plan, team member roles and workflow plan. Did you use the technology you planned on using? —TPLT]

6.2 What Went Well?

[What went well for your project management in terms of processes and technology? —TPLT]

6.3 What Went Wrong?

[What went wrong in terms of processes and technology? —TPLT]

6.4 What Would you Do Differently Next Time?

[What will you do differently for your next project? —TPLT]

7 Ethics and Equity (LO18) (Equity)

[Describe how your team applied the principle of equity and universal design to ensure equitable treatment of all stakeholders. —TPLT]

8 Reflection on Capstone

[This question focuses on what you learned during the course of the capstone project. —TPLT]

8.1 Which Courses Were Relevant

[Which of the courses you have taken were relevant for the capstone project? —TPLT]

8.2 Knowledge/Skills Outside of Courses

[What skills/knowledge did you need to acquire for your capstone project that was outside the courses you took? —TPLT]

References

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